Magnetic fields

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Interpret magnetic field patterns and compass direction.

SUCCESS CRITERIA - I can...

1. Label north and south poles.
2. Trace external field direction from north toward south.
3. Explain why field-line drawings are models.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “magnetic field” and “pole” to explain your model or result. Sentence starters are optional.

magnetic field: a field that influences magnets and moving charges

pole: a north or south region of a magnet

superposition: adding displacements or fields at the same place and time

01 Warm-up: bring back an earlier idea

A 5 Ω resistor is in series with two parallel 6 Ω resistors across 16 V.
Find parallel equivalent, total resistance and supply current.

Word help: series = connected along one unbranched current path

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $B_{net,\,x} = B_{1,\,x} + B_{2,\,x}$; $B_{net,\,y} = B_{1,\,y} + B_{2,\,y}$.
Add field components at the same point. A compass follows the net field. Outside a bar magnet, arrows run north to south.

Worked reasoning: Outside a bar magnet, field arrows run from the north pole toward the south pole. A compass's north-seeking end follows the local field direction.

Your turn: Draw curved external field lines and two small compass arrows at different points.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Label N and S on the bar magnet.
Sketch external field arrows and the direction a nearby compass would point.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

04 Apply the physics

A compass between two identical magnets has no stable preferred direction at one location.
Draw two possible opposing field contributions there.
Explain how superposition could account for the observation and why it does not prove every field is zero.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A student draws field lines that cross at one point.
Explain why this is inconsistent with a single net field direction at that point.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

In separate cases, an ideal blackbody temperature becomes 4 times as large and a sound frequency becomes 6 times as large at fixed sound speed.
Compare the two wavelength changes.
Keep the two relationships separate and label each ratio.

Word help: wavelength = distance between corresponding points on repeated waves

Sentence starter: Both … . They differ in … because … .

Case	Quantity changed	Wavelength ratio
Thermal spectrum	Temperature	____
Sound at fixed speed	Frequency	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

If a bar magnet is cut in half, do you obtain an isolated north pole and an isolated south pole?
Explain the usual result.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two magnets face N toward N.
Sketch the field qualitatively and predict the force. Then reverse one magnet and explain both changes.

Sentence starter: I predict ... . My model supports this because ... .

Label poles; add force or field arrows.



Marzano check - circle: 1 2 3